

Bocconi

COMPUTER SCIENCE

code 30424 a.y. 2024-2025

Lesson 6

Analyzing data with the spreadsheet



Università
Bocconi
MILANO

Objectives of the lesson

Understand the logics and the use of mathematical and statistical functions and pivot tables to analyze and process data in order to support decision-making processes

Our first case

We need to carry out an analysis on business software used by a group of companies available in the **Companies** worksheet:

- In a new worksheet named **Analysis** we want to calculate, for each software, the number of companies, the total number of employees and the average amount of revenues
- In the same worksheet we want to calculate:
 - how many companies have more than 100 employees
 - how many companies have revenues greater than or equal to 80,000,000 \$

Our second case

A company invested time and resources to be able to increase its presence in international tourism magazines. Data are available in the **Articles** worksheet. We need to:

- Calculate, in a new worksheet named **Press campaign**, the total number of articles with more than 5 quotations, for each year
- Calculate, in the same worksheet, the total cost on magazines with more than 25,000 copies sold, for each year
- Calculate, in the same worksheet, the average quotations on articles with cost less than or equal to 10,000 \$, for each year

Our third case

In the **Online store** worksheet we have a sample of the purchases made on the online store of a company selling computers and computer parts. We need to carry out various analyses using pivot tables, i.e. we need to calculate:

- The total purchase amount by Country and by Category
- The average quantity by Shipping and Category
- The total purchase amount by Category with the details of product codes
- The total purchase amount and the average purchase amount by year and month
- Filter the table by Country
- Group quantity by clusters
- Create a pivot chart
- ...

Tools to learn with today's lecture

- Remove Duplicates command
- COUNTIF, SUMIF, AVERAGEIF functions
- COUNTIFS, SUMIFS, AVERAGEIFS functions
- Pivot Tables and pivot charts

Book references

Excel for students in economics and finance, 2nd ed.:

Chapters 3.1.3, 3.1.4, 3.2.1, 3.2.2, 3.2.3, 3.2.4, 6.3

Excel workbook, fourth edition:

All the exercises on topics of the lesson

